In [1]:
```
import numpy as np

#Seasons
Seasons = ["2010","2011","2012","2013","2014","2015","2016","2017","2018","2019"]
Sdict = {"2010":0,"2011":1,"2012":2,"2013":3,"2014":4,"2015":5,"2016":6,"2017":7,"2

#Players
Players = ["Sachin","Rahul","Smith","Sami","Pollard","Morris","Samson","Dhoni","Koh
Pdict = {"Sachin":0,"Rahul":1,"Smith":2,"Sami":3,"Pollard":4,"Morris":5,"Samson":6,

#Salaries
Sachin_Salary = [15946875,17718750,19490625,21262500,23034375,24806250,25244493,278
Rahul_Salary = [12000000,12744189,13488377,14232567,14976754,16324500,18038573,1975
Smith_Salary = [4621800,5828090,13041250,14410581,15779912,14500000,16022500,175450
Sami_Salary = [3713640,4694041,13041250,14410581,15779912,17149243,18518574,1945000
Pollard_Salary = [4493160,4806720,6061274,13758000,15202590,16647180,18091770,19536
Morris_Salary = [3348000,4235220,12455000,14410581,15779912,14500000,16022500,17545
Samson_Salary = [3144240,3380160,3615960,4574189,13520500,14940153,16359805,1777945
Dhoni_Salary = [0,0,4171200,4484040,4796880,6053663,15506632,16669630,17832627,1899
Kohli_Salary = [0,0,0,4822800,5184480,5546160,6993708,16402500,17632688,18862875]
Sky_Salary = [3031920,3841443,13041250,14410581,15779912,14200000,15691000,17182000
#Matrix
Salary = np.array([Sachin_Salary, Rahul_Salary, Smith_Salary, Sami_Salary, Pollard_

#Games
Sachin_G = [80,77,82,82,73,82,58,78,6,35]
Rahul_G = [82,57,82,79,76,72,60,72,79,80]
Smith_G = [79,78,75,81,76,79,62,76,77,69]
Sami_G = [80,65,77,66,69,77,55,67,77,40]
Pollard_G = [82,82,82,79,82,78,54,76,71,41]
Morris_G = [70,69,67,77,70,77,57,74,79,44]
Samson_G = [78,64,80,78,45,80,60,70,62,82]
Dhoni_G = [35,35,80,74,82,78,66,81,81,27]
Kohli_G = [40,40,40,81,78,81,39,0,10,51]
Sky_G = [75,51,51,79,77,76,49,69,54,62]
#Matrix
Games = np.array([Sachin_G, Rahul_G, Smith_G, Sami_G, Pollard_G, Morris_G, Samson_G

#Points
Sachin_PTS = [2832,2430,2323,2201,1970,2078,1616,2133,83,782]
Rahul_PTS = [1653,1426,1779,1688,1619,1312,1129,1170,1245,1154]
Smith_PTS = [2478,2132,2250,2304,2258,2111,1683,2036,2089,1743]
Sami_PTS = [2122,1881,1978,1504,1943,1970,1245,1920,2112,966]
Pollard_PTS = [1292,1443,1695,1624,1503,1784,1113,1296,1297,646]
Morris_PTS = [1572,1561,1496,1746,1678,1438,1025,1232,1281,928]
Samson_PTS = [1258,1104,1684,1781,841,1268,1189,1186,1185,1564]
Dhoni_PTS = [903,903,1624,1871,2472,2161,1850,2280,2593,686]
Kohli_PTS = [597,597,597,1361,1619,2026,852,0,159,904]
Sky_PTS = [2040,1397,1254,2386,2045,1941,1082,1463,1028,1331]
#Matrix
Points = np.array([Sachin_PTS, Rahul_PTS, Smith_PTS, Sami_PTS, Pollard_PTS, Morris_
```

In [2]:
```
Games
```

Out[2]:
```
array([[80, 77, 82, 82, 73, 82, 58, 78,  6, 35],
       [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
       [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
       [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
       [82, 82, 82, 79, 82, 78, 54, 76, 71, 41],
       [70, 69, 67, 77, 70, 77, 57, 74, 79, 44],
       [78, 64, 80, 78, 45, 80, 60, 70, 62, 82],
       [35, 35, 80, 74, 82, 78, 66, 81, 81, 27],
       [40, 40, 40, 81, 78, 81, 39,  0, 10, 51],
       [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]])
```

In [3]:
```
Salary
```

Out[3]:
```
array([[15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
        25244493, 27849149, 30453805, 23500000],
       [12000000, 12744189, 13488377, 14232567, 14976754, 16324500,
        18038573, 19752645, 21466718, 23180790],
       [ 4621800,  5828090, 13041250, 14410581, 15779912, 14500000,
        16022500, 17545000, 19067500, 20644400],
       [ 3713640,  4694041, 13041250, 14410581, 15779912, 17149243,
        18518574, 19450000, 22407474, 22458000],
       [ 4493160,  4806720,  6061274, 13758000, 15202590, 16647180,
        18091770, 19536360, 20513178, 21436271],
       [ 3348000,  4235220, 12455000, 14410581, 15779912, 14500000,
        16022500, 17545000, 19067500, 20644400],
       [ 3144240,  3380160,  3615960,  4574189, 13520500, 14940153,
        16359805, 17779458, 18668431, 20068563],
       [       0,        0,  4171200,  4484040,  4796880,  6053663,
        15506632, 16669630, 17832627, 18995624],
       [       0,        0,        0,  4822800,  5184480,  5546160,
         6993708, 16402500, 17632688, 18862875],
       [ 3031920,  3841443, 13041250, 14410581, 15779912, 14200000,
        15691000, 17182000, 18673000, 15000000]])
```

In [8]:
```
Pdict
```

Out[8]:
```
{'Sachin': 0,
 'Rahul': 1,
 'Smith': 2,
 'Sami': 3,
 'Pollard': 4,
 'Morris': 5,
 'Samson': 6,
 'Dhoni': 7,
 'Kohli': 8,
 'Sky': 9}
```

In [9]:
```
Sdict
```

Out[9]:
```
{'2010': 0,
 '2011': 1,
 '2012': 2,
 '2013': 3,
 '2014': 4,
 '2015': 5,
 '2016': 6,
 '2017': 7,
 '2018': 8,
 '2019': 9}
```

In [7]:
```
Salary/Games
```

```
C:\Users\srini\AppData\Local\Temp\ipykernel_26864\3709746658.py:1: RuntimeWarning: d
ivide by zero encountered in divide
  Salary/Games
```

Out[7]:
```
array([[ 199335.9375    ,  230113.63636364,  237690.54878049,
         259298.7804878 ,  315539.38356164,  302515.24390244,
         435249.87931034,  357040.37179487, 5075634.16666667,
         671428.57142857],
       [ 146341.46341463,  223582.26315789,  164492.40243902,
         180159.07594937,  197062.55263158,  226729.16666667,
         300642.88333333,  274342.29166667,  271730.60759494,
         289759.875     ],
       [  58503.79746835,   74719.1025641 ,  173883.33333333,
         177908.40740741,  207630.42105263,  183544.30379747,
         258427.41935484,  230855.26315789,  247629.87012987,
         299194.20289855],
       [  46420.5       ,   72216.01538462,  169366.88311688,
         218342.13636364,  228694.37681159,  222717.44155844,
         336701.34545455,  290298.50746269,  291006.15584416,
         561450.        ],
       [  54794.63414634,   58618.53658537,   73917.97560976,
         174151.89873418,  185397.43902439,  213425.38461538,
         335032.77777778,  257057.36842105,  288918.        ,
         522835.87804878],
       [  47828.57142857,   61380.        ,  185895.52238806,
         187150.4025974 ,  225427.31428571,  188311.68831169,
         281096.49122807,  237094.59459459,  241360.75949367,
         469190.90909091],
       [  40310.76923077,   52815.        ,   45199.5       ,
          58643.44871795,  300455.55555556,  186751.9125    ,
         272663.41666667,  253992.25714286,  301103.72580645,
         244738.57317073],
       [      0.        ,       0.        ,   52140.        ,
          60595.13513514,   58498.53658537,   77611.06410256,
         234948.96969697,  205797.90123457,  220155.88888889,
         703541.62962963],
       [      0.        ,       0.        ,       0.        ,
          59540.74074074,   66467.69230769,   68471.11111111,
         179325.84615385,              inf, 1763268.8       ,
         369860.29411765],
       [  40425.6       ,   75322.41176471,  255710.78431373,
         182412.41772152,  204933.92207792,  186842.10526316,
         320224.48979592,  249014.49275362,  345796.2962963 ,
         241935.48387097]])
```

In [10]:
```
np.round(Salary//Games)
```

```
C:\Users\srini\AppData\Local\Temp\ipykernel_26864\3663165759.py:1: RuntimeWarning: d
ivide by zero encountered in floor_divide
  np.round(Salary//Games)
```

Out[10]:
```
array([[ 199335,  230113,  237690,  259298,  315539,  302515,  435249,
         357040, 5075634,  671428],
       [ 146341,  223582,  164492,  180159,  197062,  226729,  300642,
         274342,  271730,  289759],
       [  58503,   74719,  173883,  177908,  207630,  183544,  258427,
         230855,  247629,  299194],
       [  46420,   72216,  169366,  218342,  228694,  222717,  336701,
         290298,  291006,  561450],
       [  54794,   58618,   73917,  174151,  185397,  213425,  335032,
         257057,  288918,  522835],
       [  47828,   61380,  185895,  187150,  225427,  188311,  281096,
         237094,  241360,  469190],
       [  40310,   52815,   45199,   58643,  300455,  186751,  272663,
         253992,  301103,  244738],
       [      0,       0,   52140,   60595,   58498,   77611,  234948,
         205797,  220155,  703541],
       [      0,       0,       0,   59540,   66467,   68471,  179325,
               0, 1763268,  369860],
       [  40425,   75322,  255710,  182412,  204933,  186842,  320224,
         249014,  345796,  241935]])
```

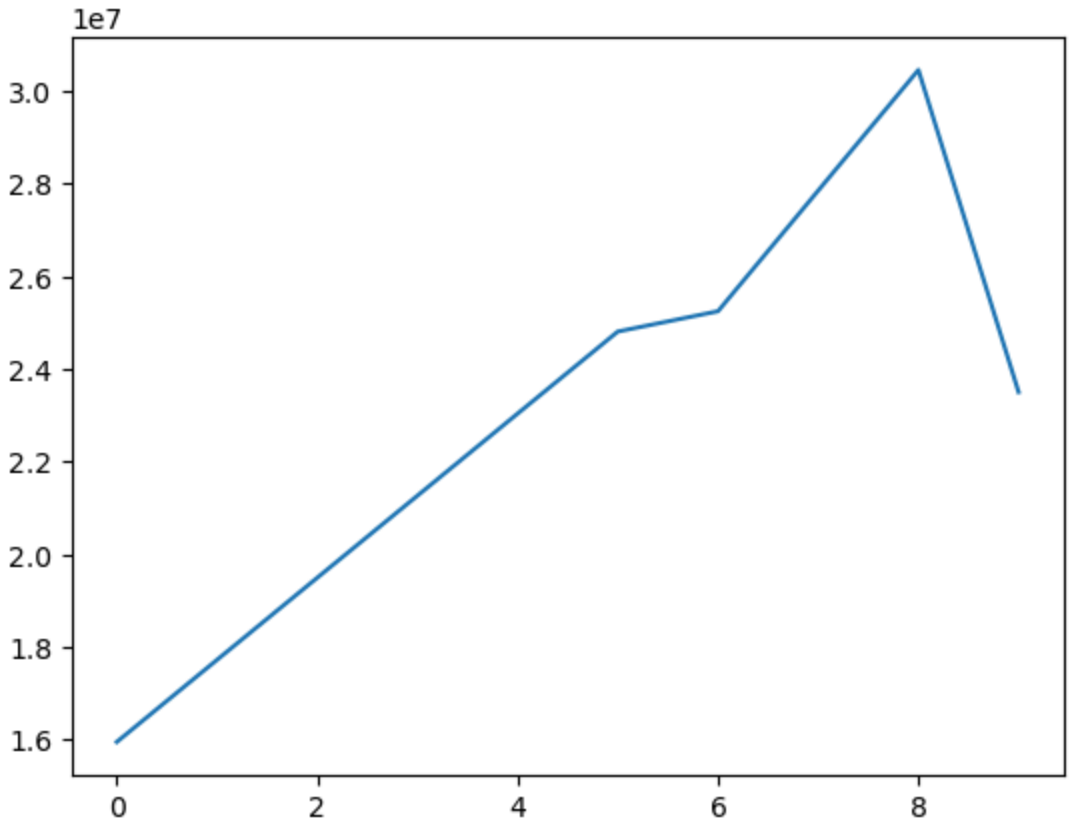
In [24]:
```
import matplotlib.pyplot as plt
```

In [25]:
```
Salary[0]
```

Out[25]:
```
array([15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
       25244493, 27849149, 30453805, 23500000])
```

In [26]:
```
plt.plot(Salary[0])
```

Out[26]:
```
[<matplotlib.lines.Line2D at 0x1e034bbe0d0>]
```

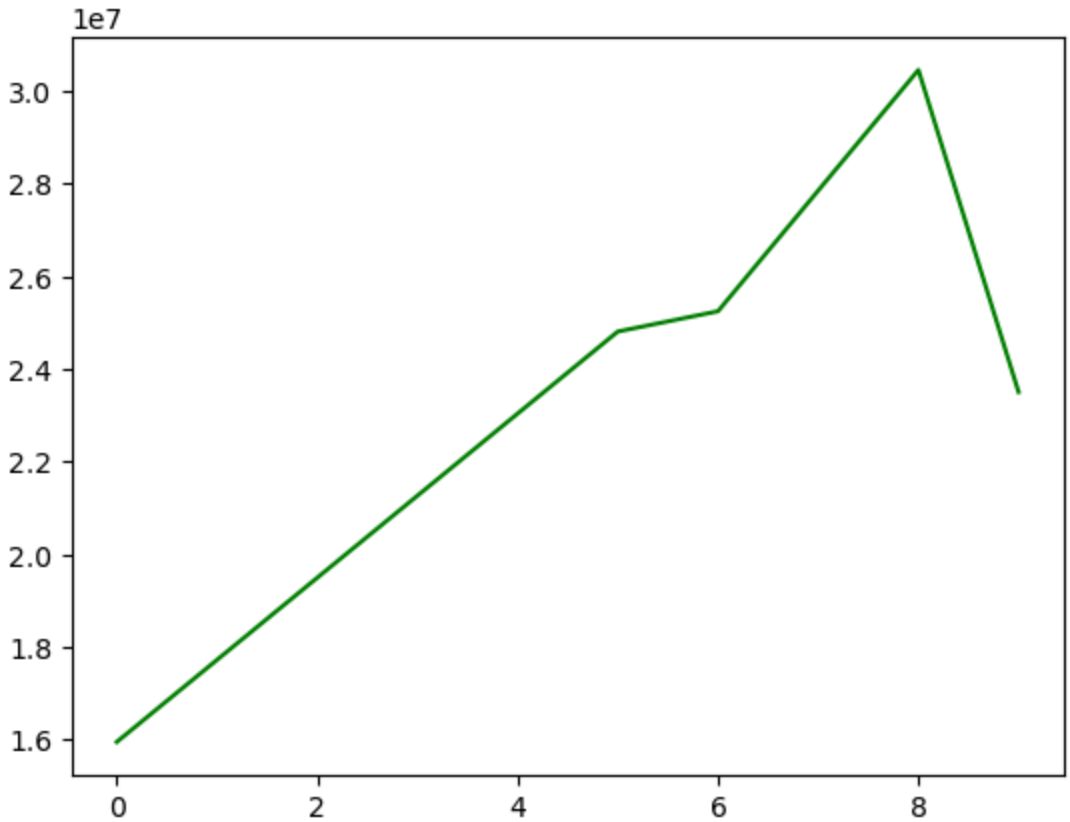


In [27]:
```
Games[0]
```

Out[27]:
```
array([80, 77, 82, 82, 73, 82, 58, 78,  6, 35])
```

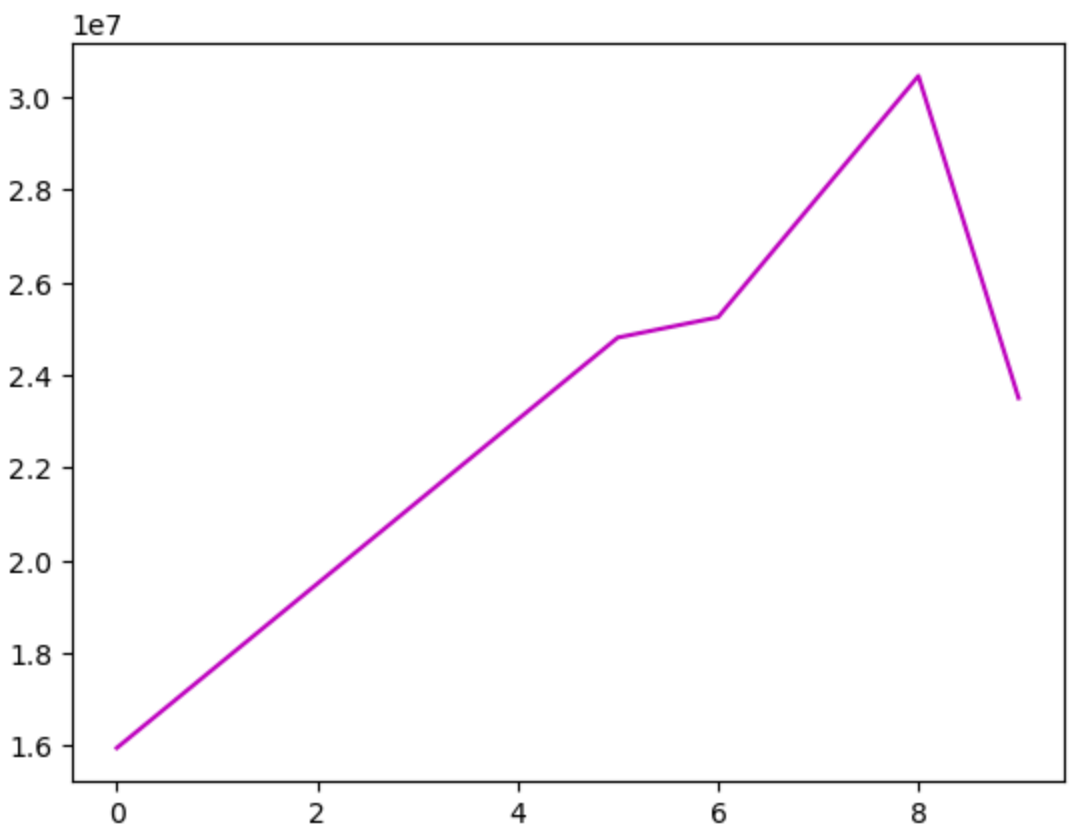
In [29]:
```
plt.plot(Salary[0], color='green')
```

Out[29]:
```
[<matplotlib.lines.Line2D at 0x1e034ac4cd0>]
```



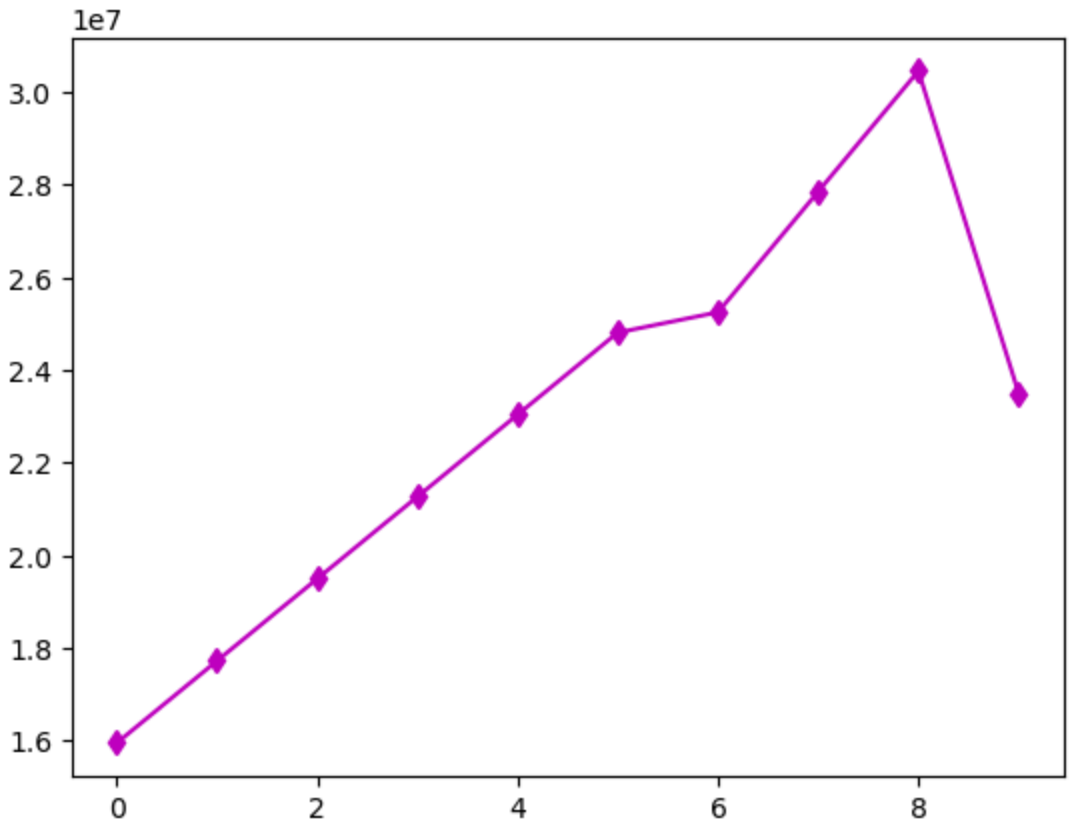
In [32]:
```
plt.plot(Salary[0], c='m')
```

Out[32]: [<matplotlib.lines.Line2D at 0x1e034d07b10>]



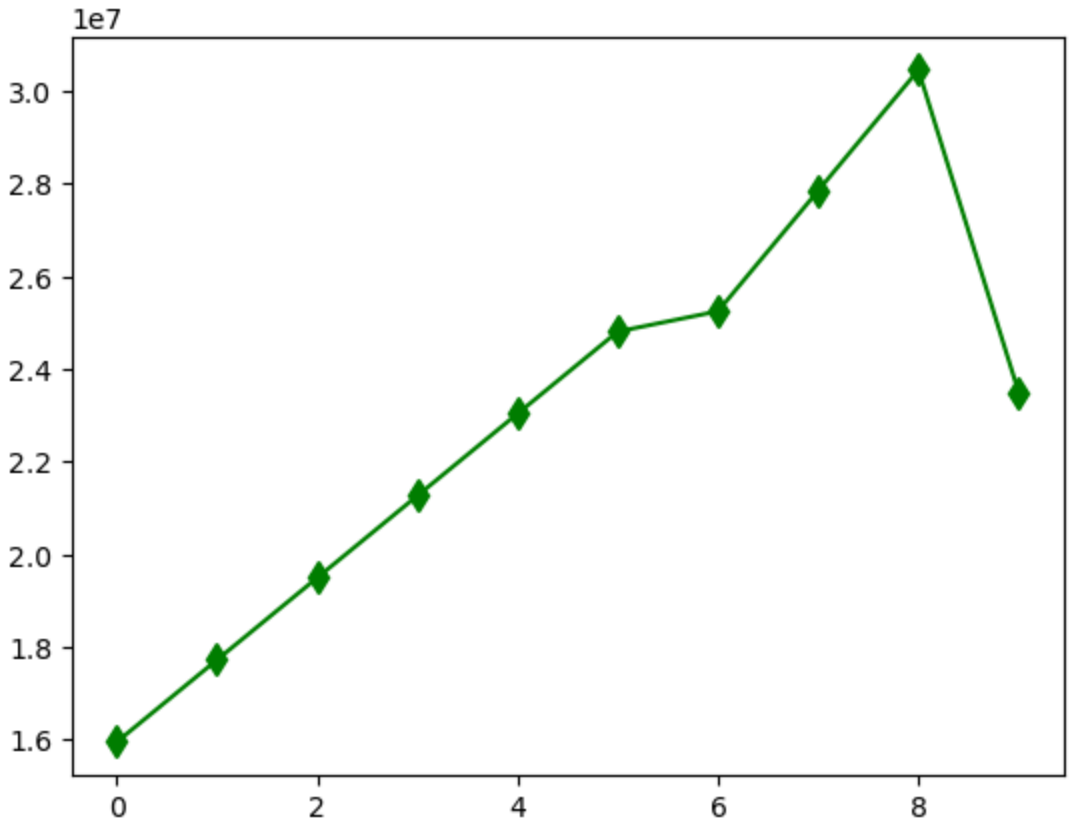
In [35]:
```
plt.plot(Salary[0], c='m', marker='d')
```

Out[35]:
```
[<matplotlib.lines.Line2D at 0x1e034f5ed50>]
```



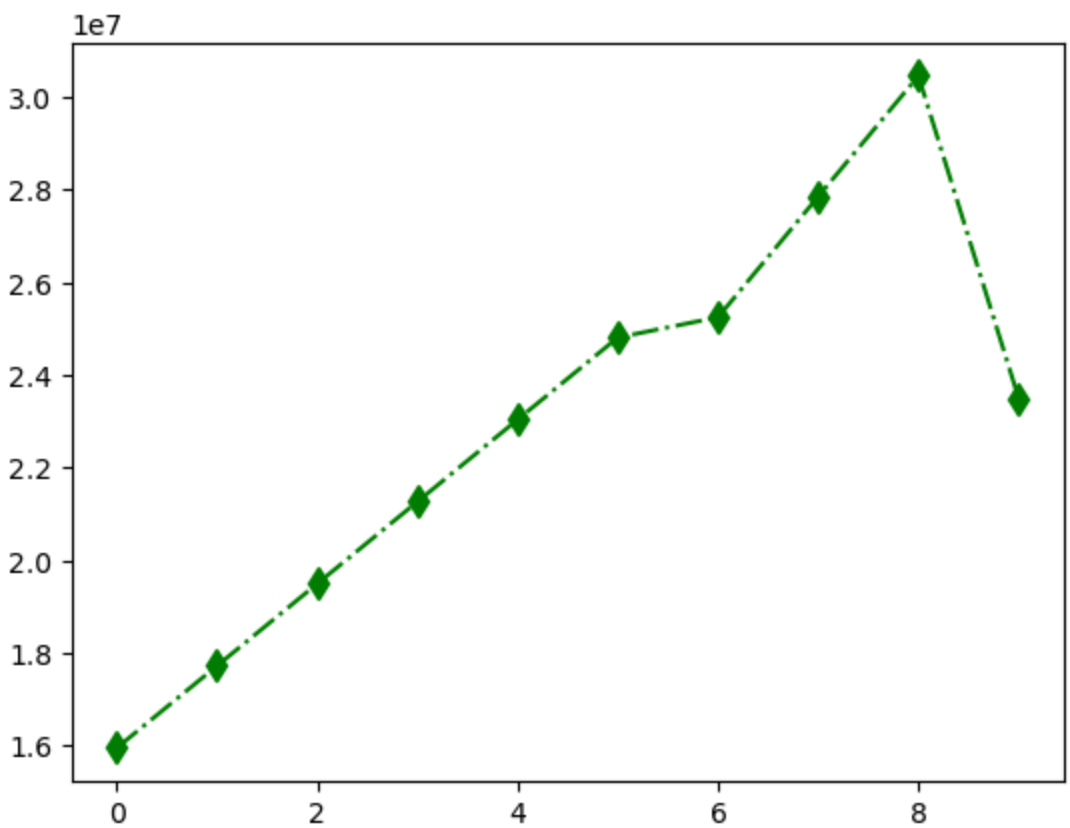
In [40]:
```
plt.plot(Salary[0], c='g', marker='d', ms=8)
```

Out[40]:
```
[<matplotlib.lines.Line2D at 0x1e036066710>]
```



In [46]:
```
plt.plot(Salary[0], c='g', marker='d', ms=8, ls='-.')
```

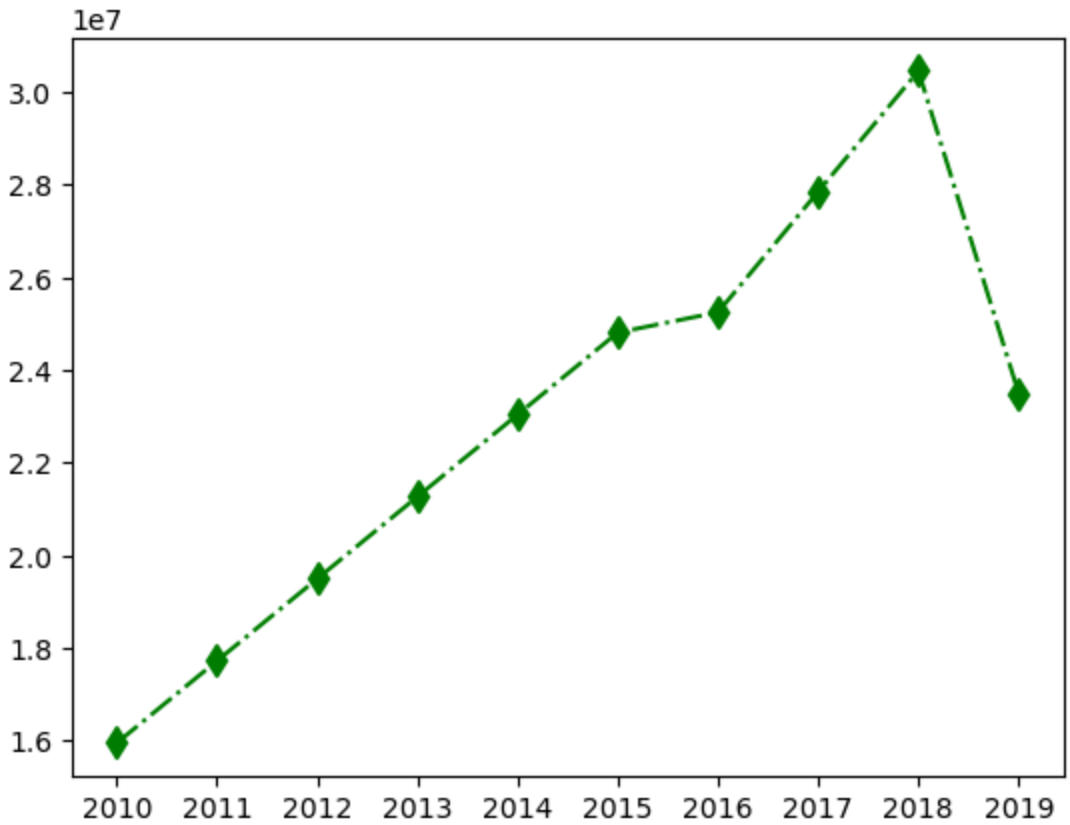
Out[46]: [<matplotlib.lines.Line2D at 0x1e0363e6210>]



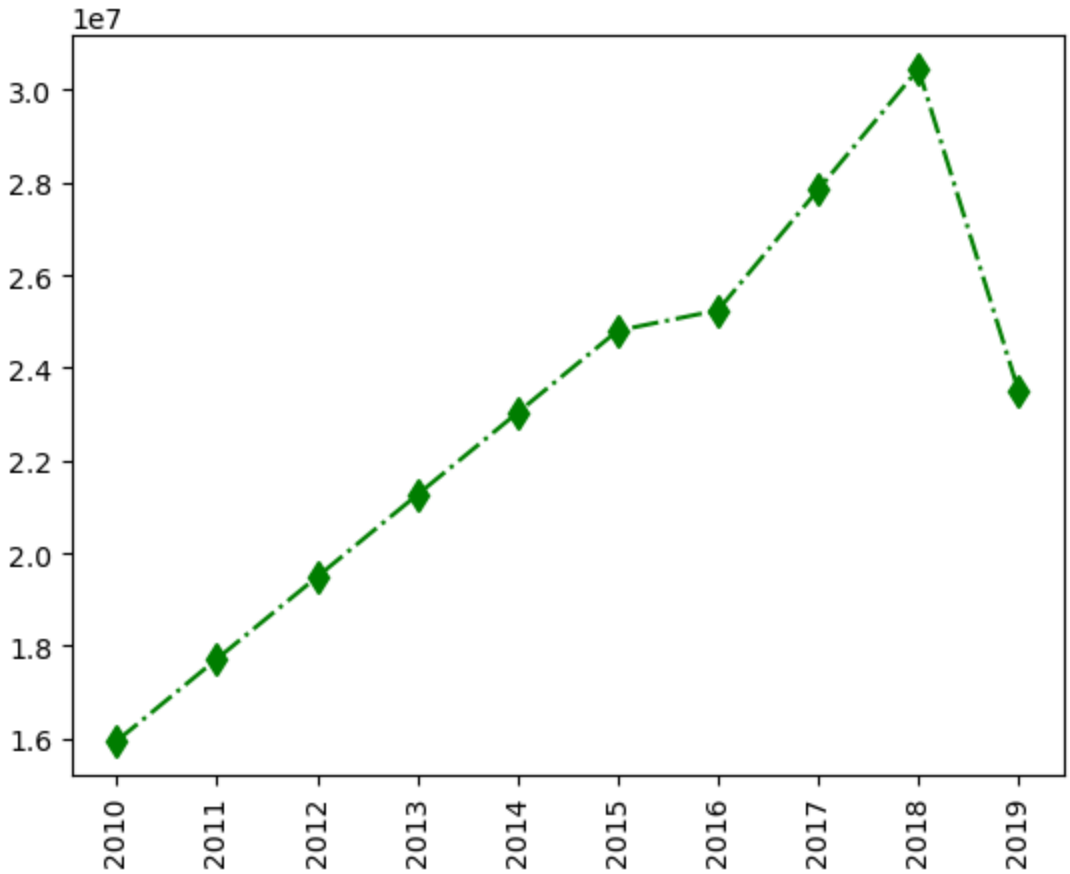
In [47]:
```
Sdict
```

Out[47]:
```
{'2010': 0,
 '2011': 1,
 '2012': 2,
 '2013': 3,
 '2014': 4,
 '2015': 5,
 '2016': 6,
 '2017': 7,
 '2018': 8,
 '2019': 9}
```

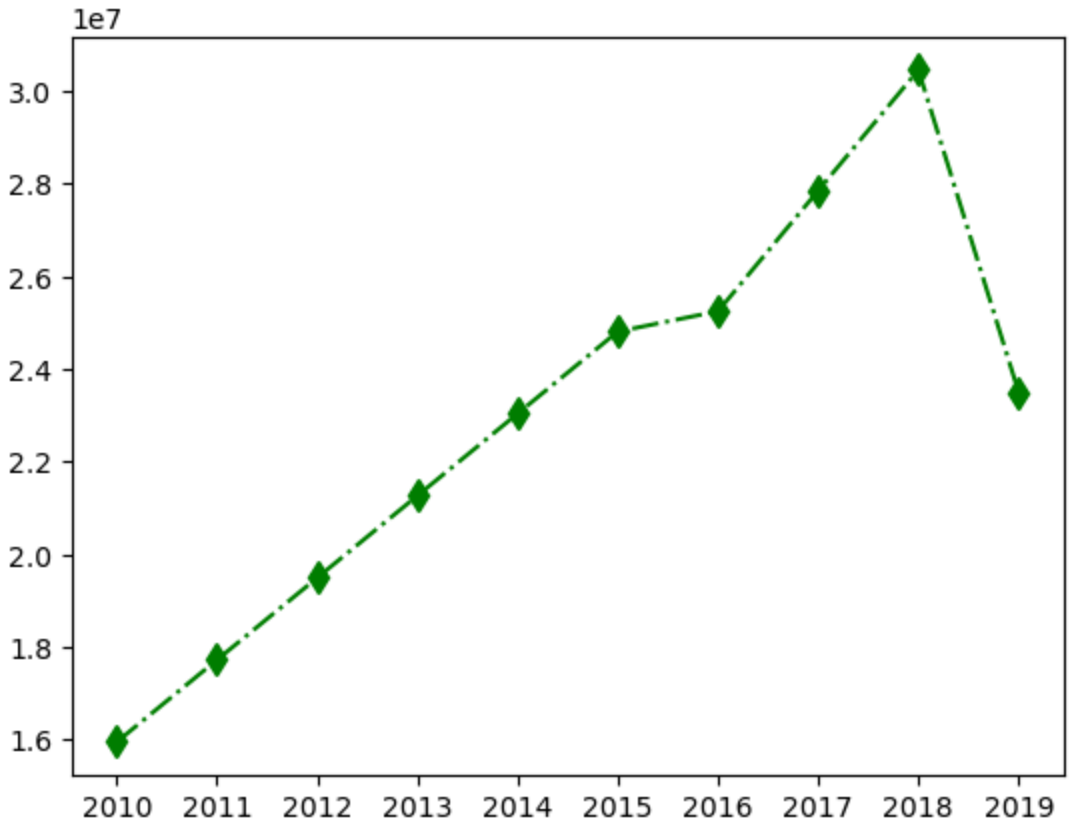
In [48]:
```
plt.plot(Salary[0], c='g', marker='d', ms=8, ls='-.')
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```



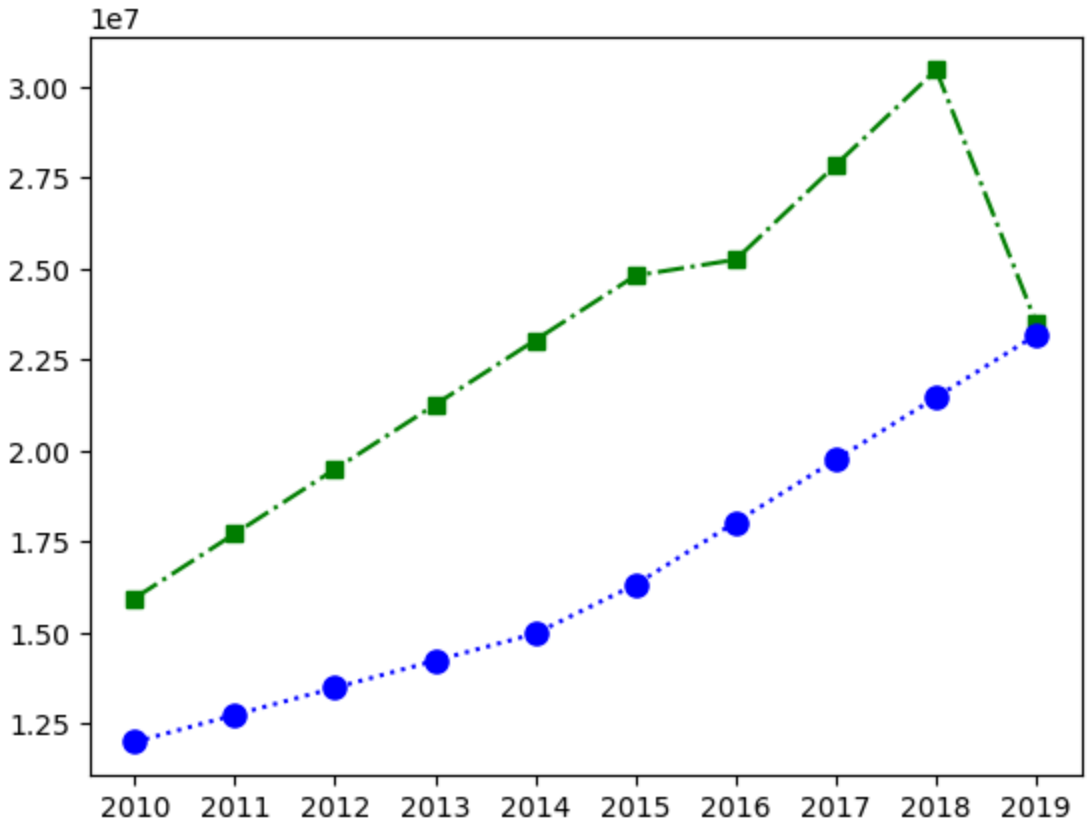
In [49]:
```
plt.plot(Salary[0], c='g', marker='d', ms=8, ls='-.')
plt.xticks(list(range(0,10)),Seasons, rotation='vertical')
plt.show()
```



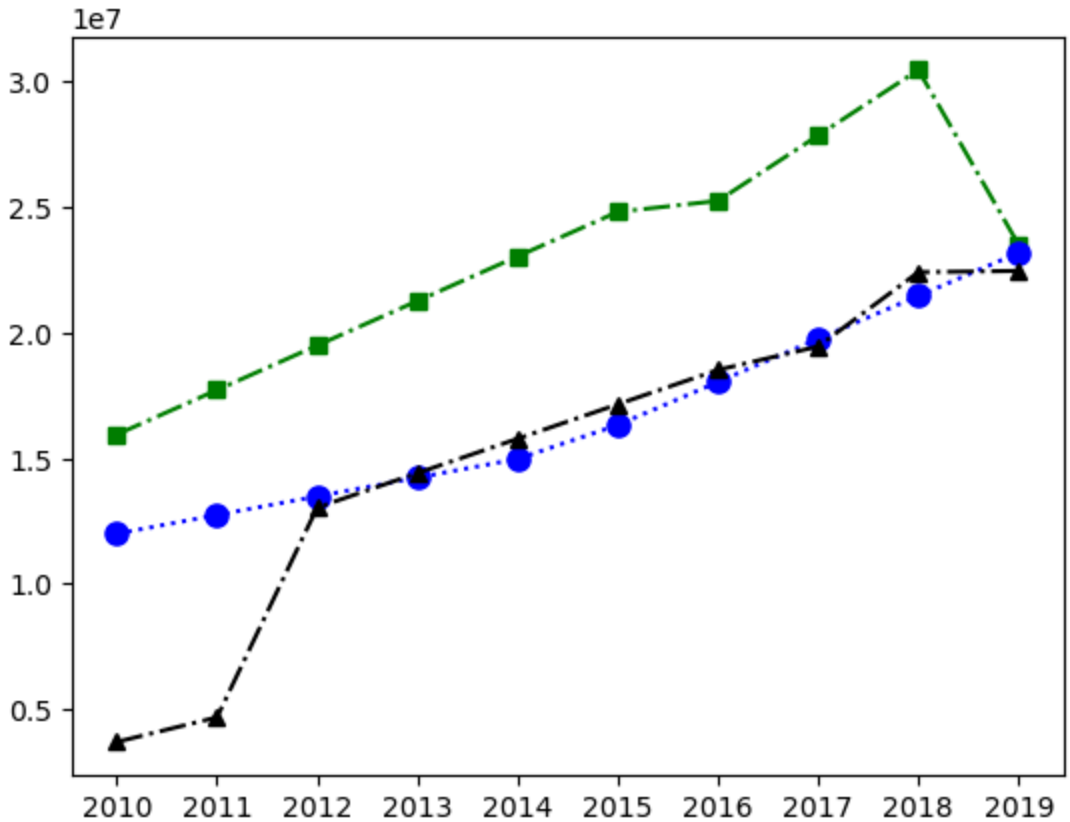
In [50]:
```
plt.plot(Salary[0], c='g', marker='d', ms=8, ls='-.')
plt.xticks(list(range(0,10)),Seasons,rotation='horizontal')
plt.show()
```



In [54]:
```
plt.plot(Salary[0], c='g', marker='s', ms=6, ls='-.')
plt.plot(Salary[1], c='b', marker='o', ms=8, ls=':')
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```

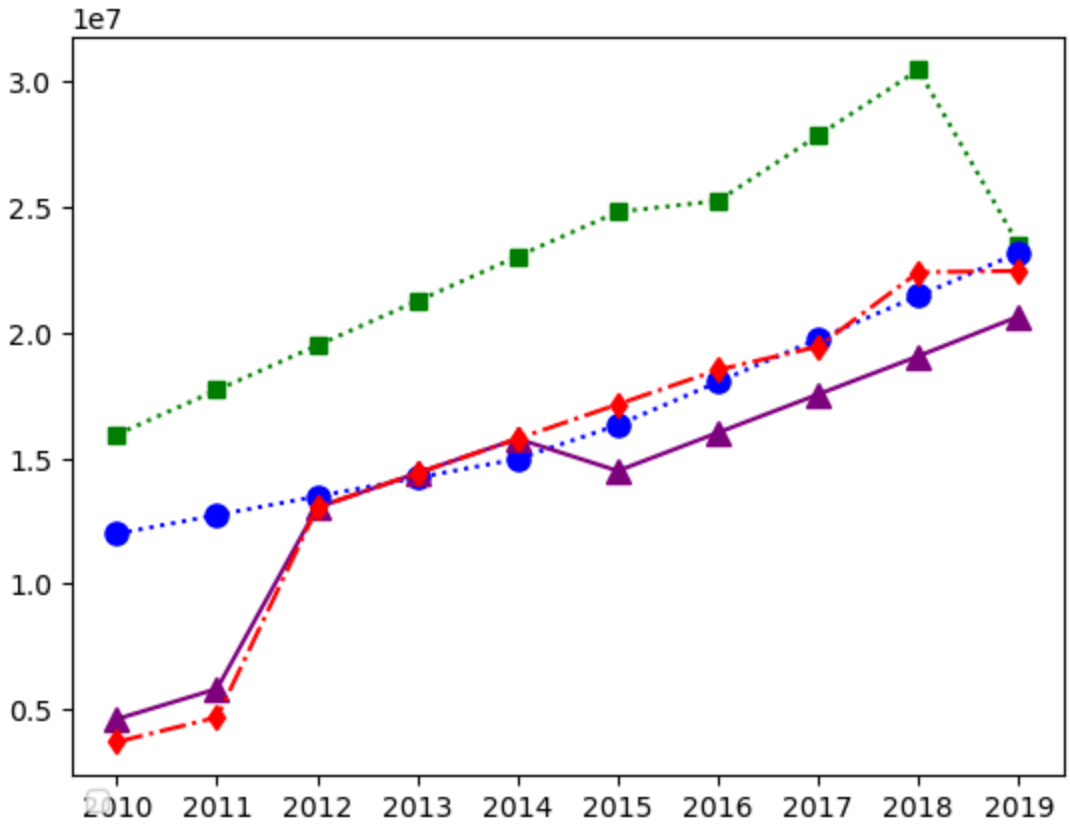


In [56]:
```
plt.plot(Salary[0], c='g', marker='s', ms=6, ls='-.')
plt.plot(Salary[1], c='b', marker='o', ms=8, ls=':')
plt.plot(Salary[3], c='k', marker='^', ms=6, ls='-.')
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```

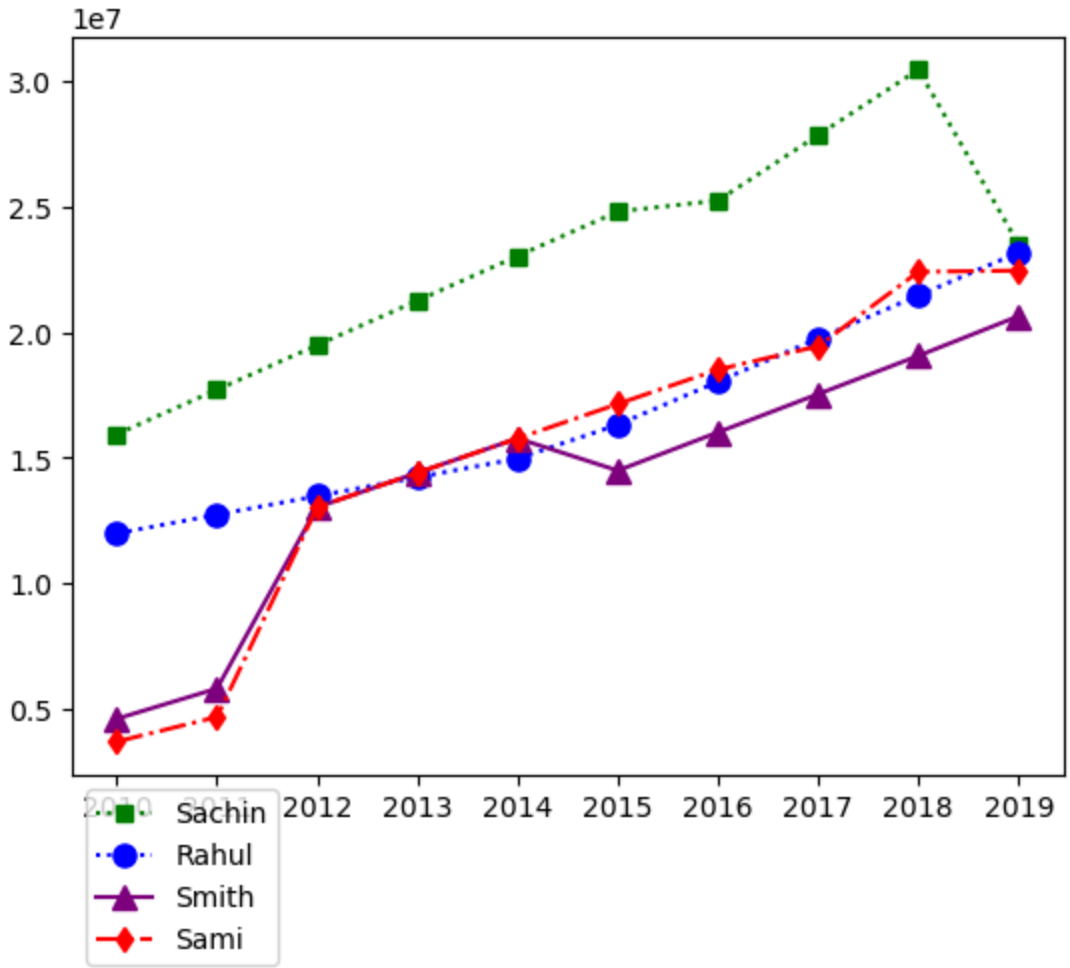


In [62]:
```
plt.plot(Salary[0], c='g', marker='s', ms=6, ls=':')
plt.plot(Salary[1], c='b', marker='o', ms=8, ls=':')
plt.plot(Salary[2], c='purple', marker='^', ms=8, ls='-')
plt.plot(Salary[3], c='r', marker='d', ms=6, ls='-.')
plt.legend(loc='upper left',bbox_to_anchor=(0,0))
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```

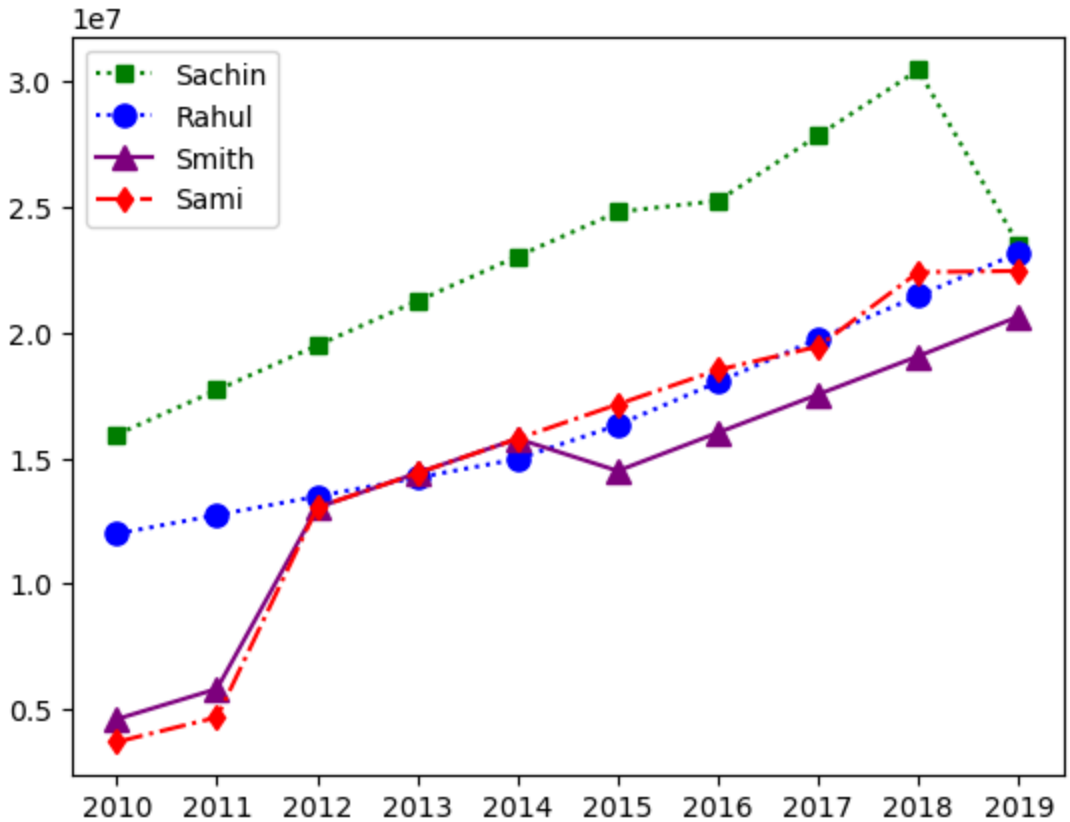
```
C:\Users\srini\AppData\Local\Temp\ipykernel_26864\2252571047.py:5: UserWarning: No a
rtists with labels found to put in legend.  Note that artists whose label start with
an underscore are ignored when legend() is called with no argument.
  plt.legend(loc='upper left',bbox_to_anchor=(0,0))
```



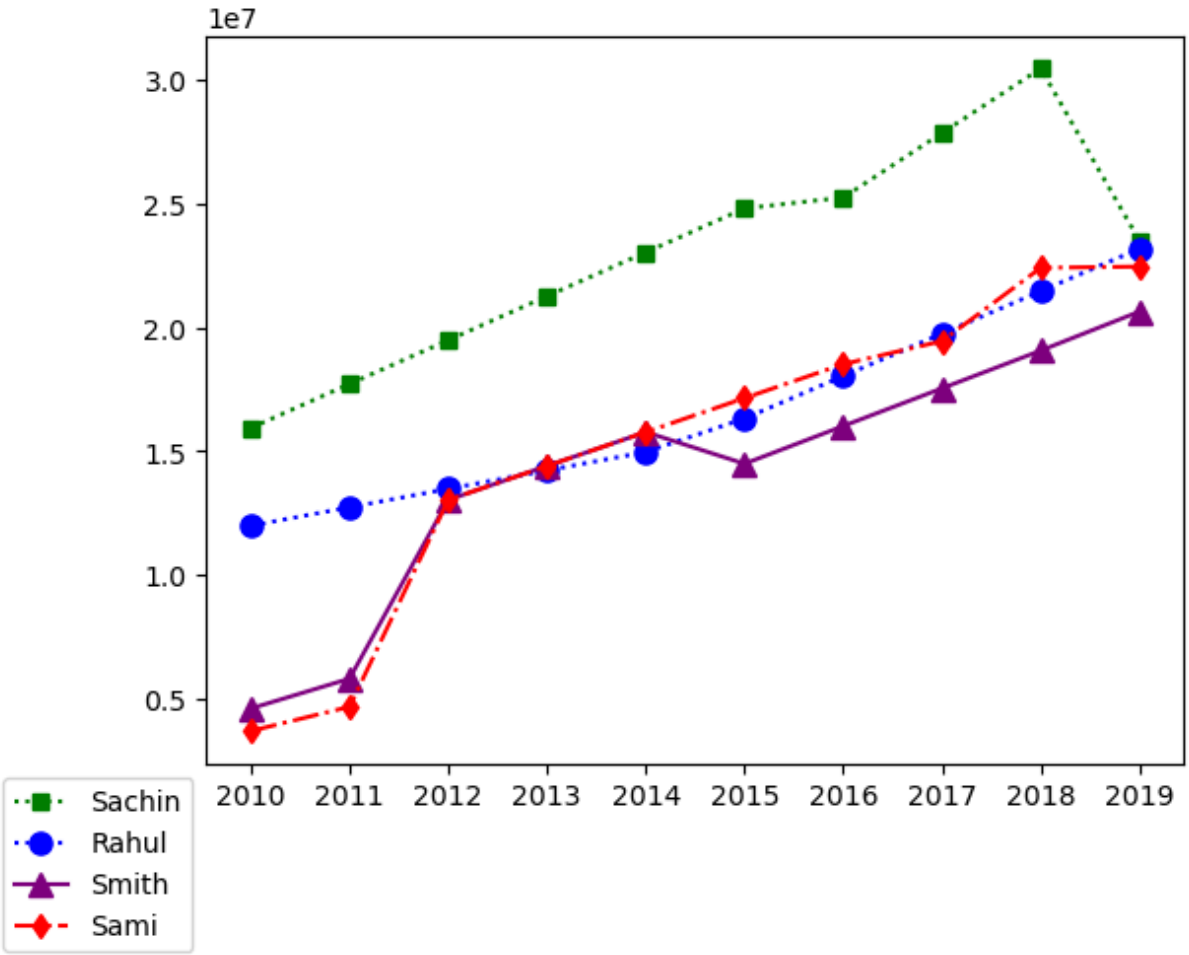
In [65]:
```
plt.plot(Salary[0], c='g', marker='s', ms=6, ls=':',label=Players[0])
plt.plot(Salary[1], c='b', marker='o', ms=8, ls=':',label=Players[1])
plt.plot(Salary[2], c='purple', marker='^', ms=8, ls='-',label=Players[2])
plt.plot(Salary[3], c='r', marker='d', ms=6, ls='-.',label=Players[3])
plt.legend(loc='upper left',bbox_to_anchor=(0,0))
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```



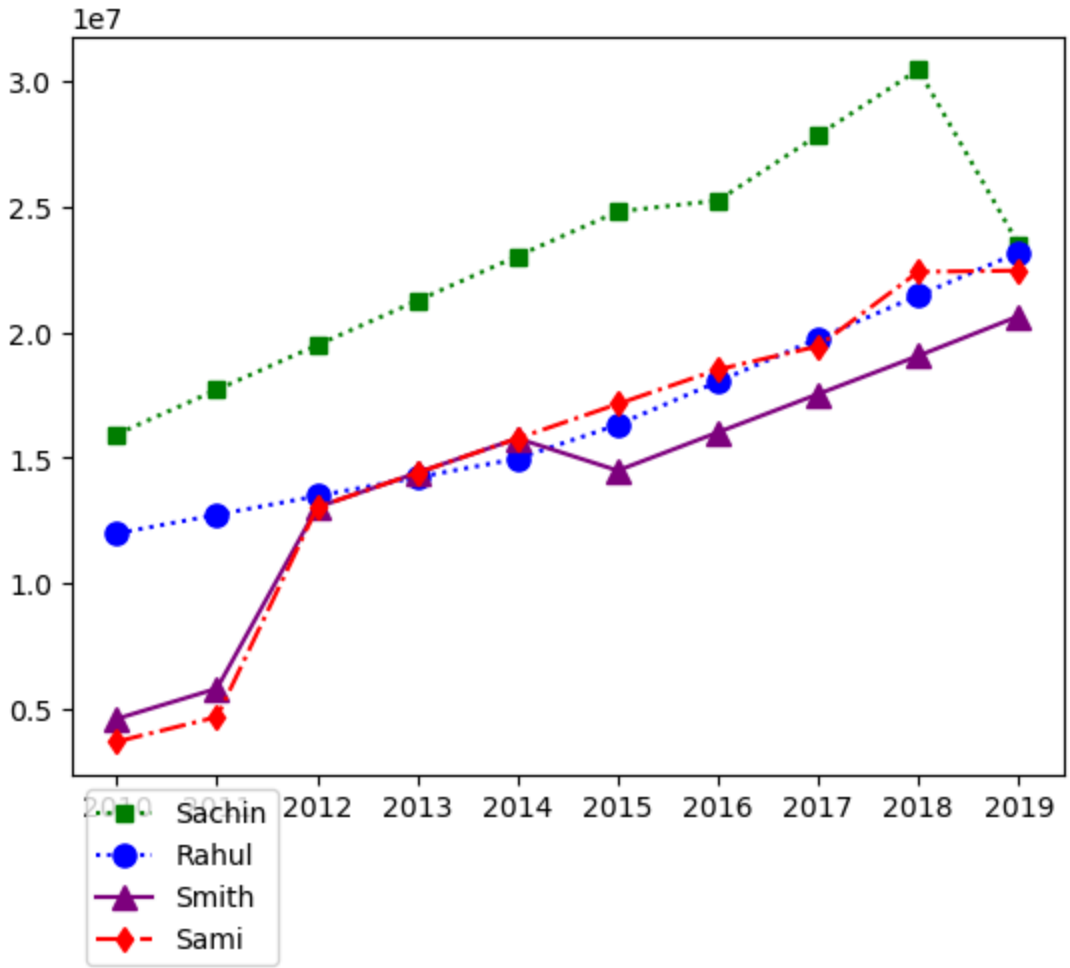
In [66]:
```
plt.plot(Salary[0], c='g', marker='s', ms=6, ls=':',label=Players[0])
plt.plot(Salary[1], c='b', marker='o', ms=8, ls=':',label=Players[1])
plt.plot(Salary[2], c='purple', marker='^', ms=8, ls='-',label=Players[2])
plt.plot(Salary[3], c='r', marker='d', ms=6, ls='-.',label=Players[3])
plt.legend()
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```



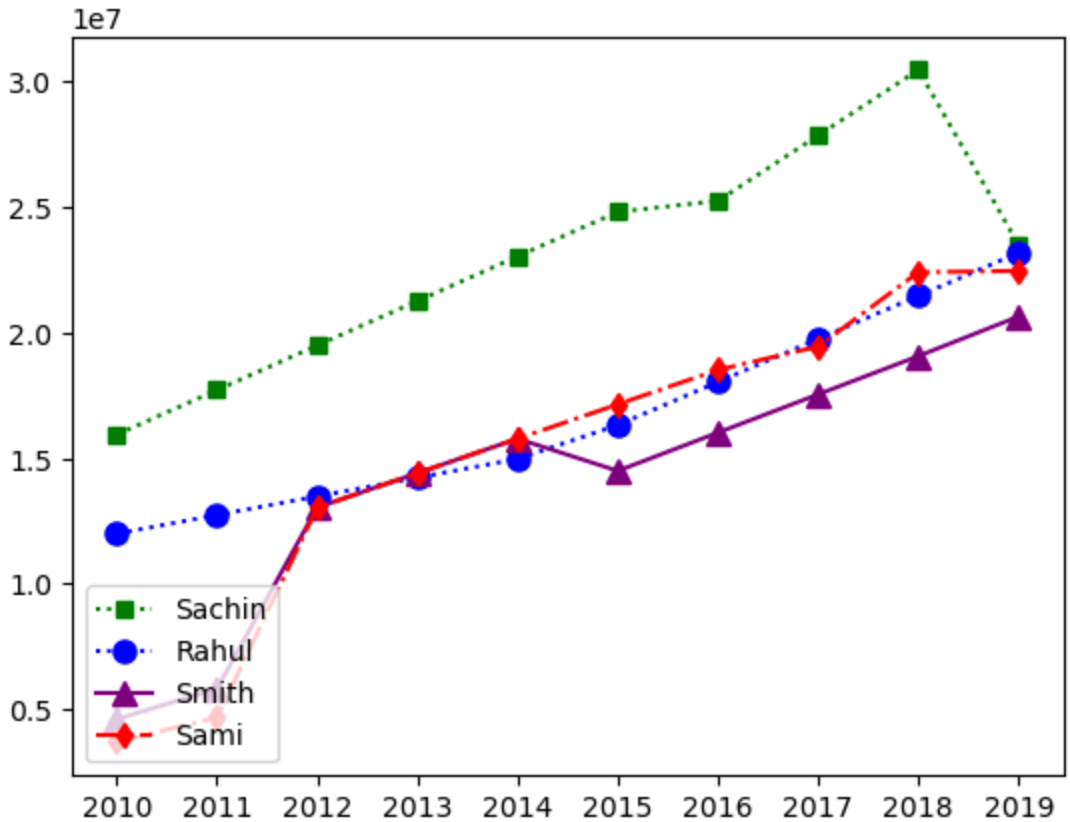
In [71]:
```
plt.plot(Salary[0], c='g', marker='s', ms=6, ls=':',label=Players[0])
plt.plot(Salary[1], c='b', marker='o', ms=8, ls=':',label=Players[1])
plt.plot(Salary[2], c='purple', marker='^', ms=8, ls='-',label=Players[2])
plt.plot(Salary[3], c='r', marker='d', ms=6, ls='-.',label=Players[3])
plt.legend(loc='upper right',bbox_to_anchor=(0,0))
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```



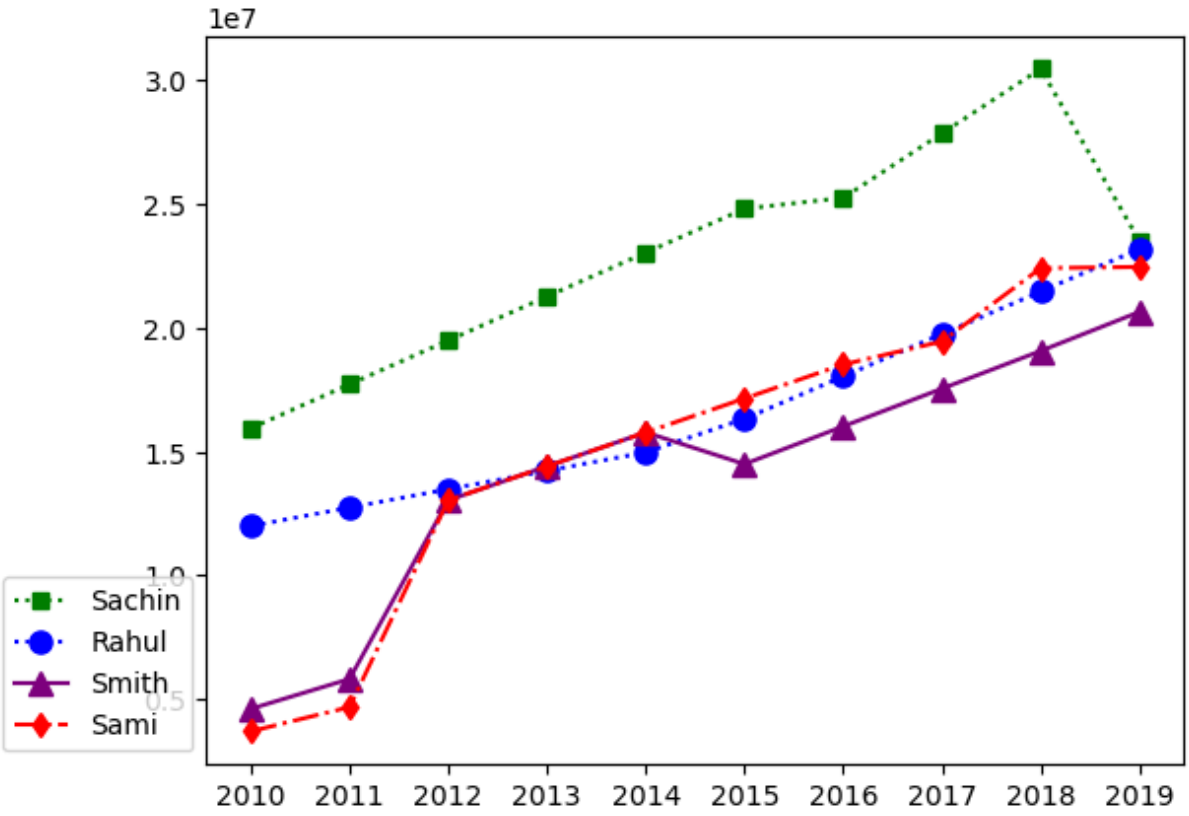
In [72]:
```
plt.plot(Salary[0], c='g', marker='s', ms=6, ls=':',label=Players[0])
plt.plot(Salary[1], c='b', marker='o', ms=8, ls=':',label=Players[1])
plt.plot(Salary[2], c='purple', marker='^', ms=8, ls='-',label=Players[2])
plt.plot(Salary[3], c='r', marker='d', ms=6, ls='-.',label=Players[3])
plt.legend(loc='upper left',bbox_to_anchor=(0,0))
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```



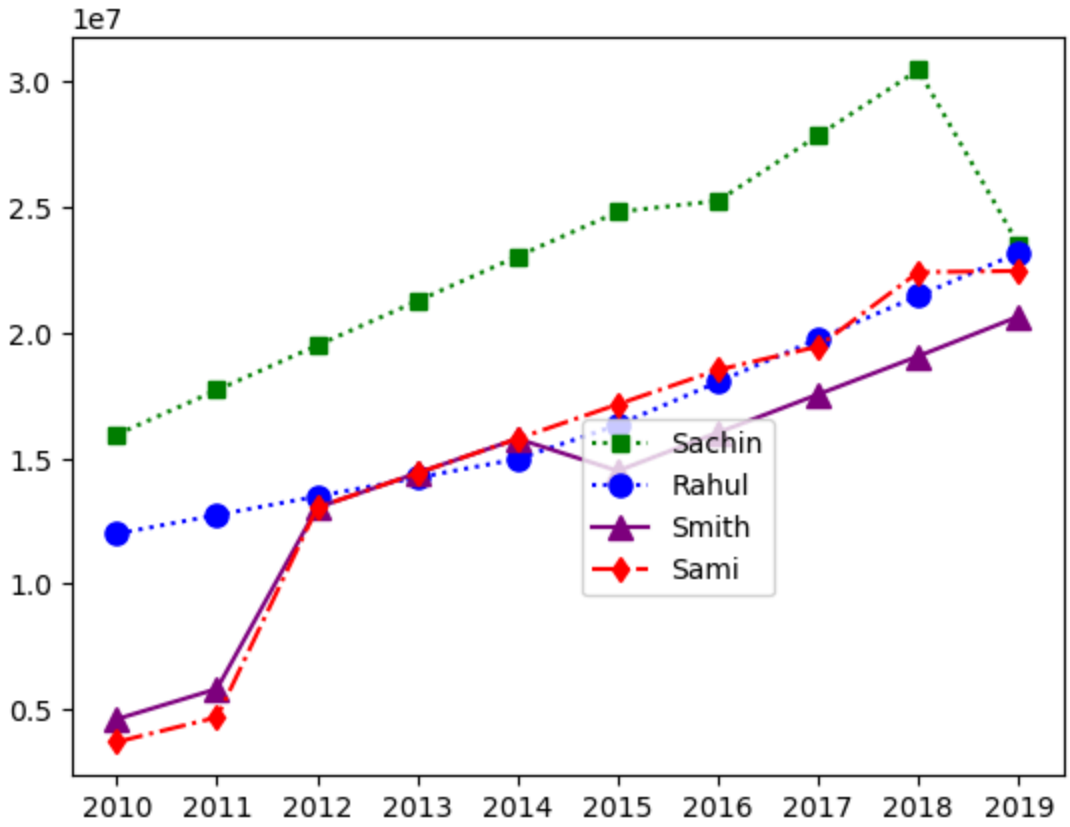
In [73]:
```
plt.plot(Salary[0], c='g', marker='s', ms=6, ls=':',label=Players[0])
plt.plot(Salary[1], c='b', marker='o', ms=8, ls=':',label=Players[1])
plt.plot(Salary[2], c='purple', marker='^', ms=8, ls='-',label=Players[2])
plt.plot(Salary[3], c='r', marker='d', ms=6, ls='-.',label=Players[3])
plt.legend(loc='lower left',bbox_to_anchor=(0,0))
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```



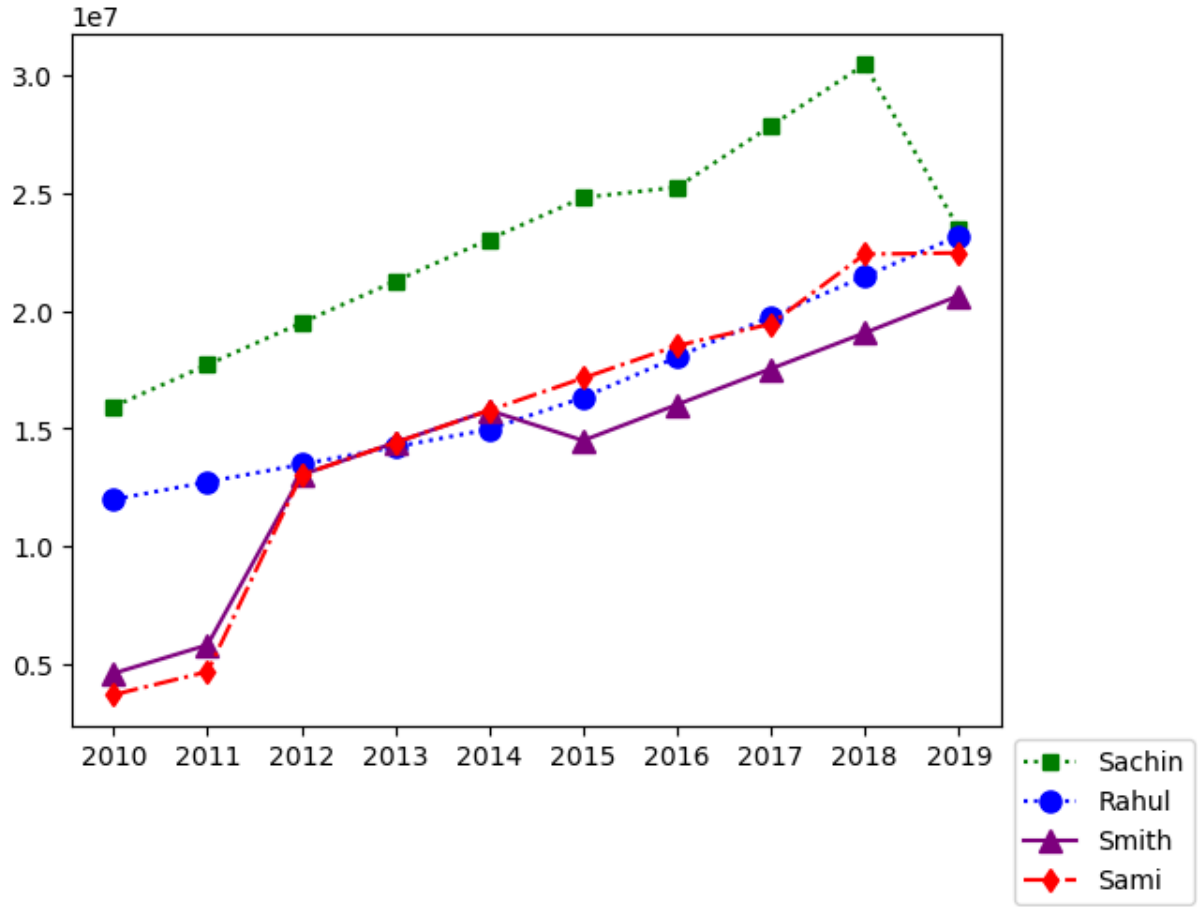
In [75]:
```
plt.plot(Salary[0], c='g', marker='s', ms=6, ls=':',label=Players[0])
plt.plot(Salary[1], c='b', marker='o', ms=8, ls=':',label=Players[1])
plt.plot(Salary[2], c='purple', marker='^', ms=8, ls='-',label=Players[2])
plt.plot(Salary[3], c='r', marker='d', ms=6, ls='-.',label=Players[3])
plt.legend(loc='lower right',bbox_to_anchor=(0,0))
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```



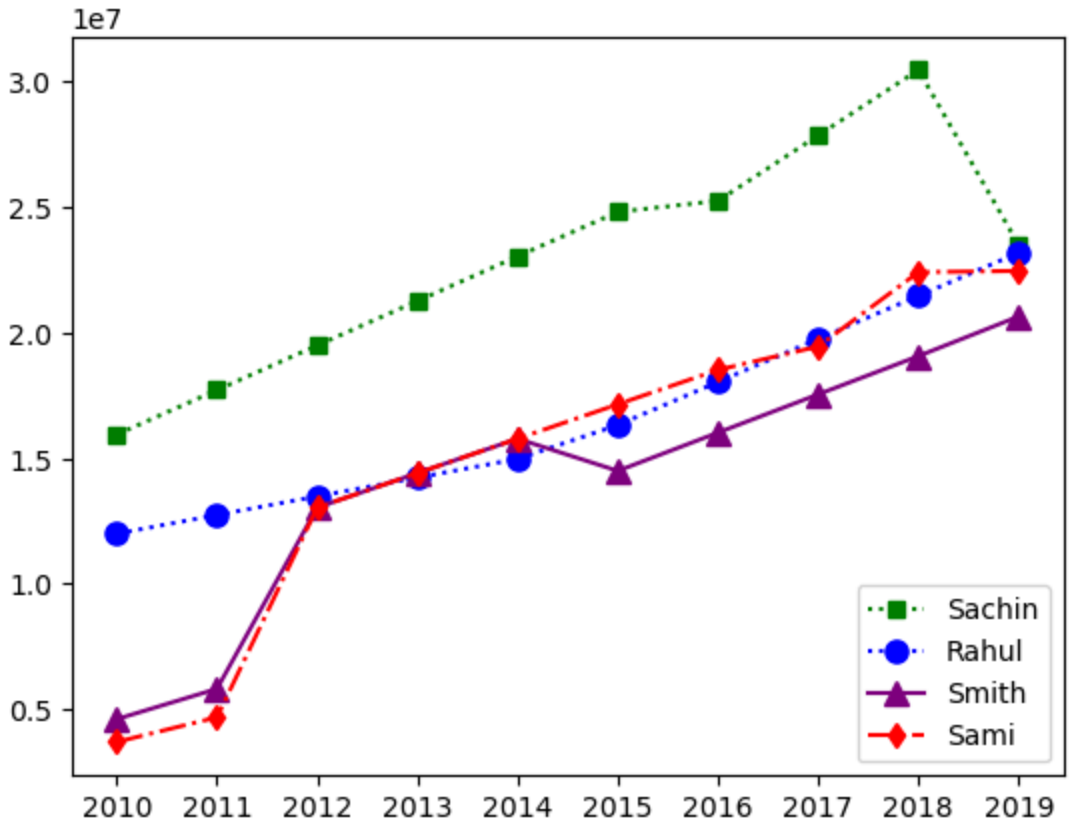
In [76]:
```
plt.plot(Salary[0], c='g', marker='s', ms=6, ls=':',label=Players[0])
plt.plot(Salary[1], c='b', marker='o', ms=8, ls=':',label=Players[1])
plt.plot(Salary[2], c='purple', marker='^', ms=8, ls='-',label=Players[2])
plt.plot(Salary[3], c='r', marker='d', ms=6, ls='-.',label=Players[3])
plt.legend(loc='upper left',bbox_to_anchor=(0.5,0.5))
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```



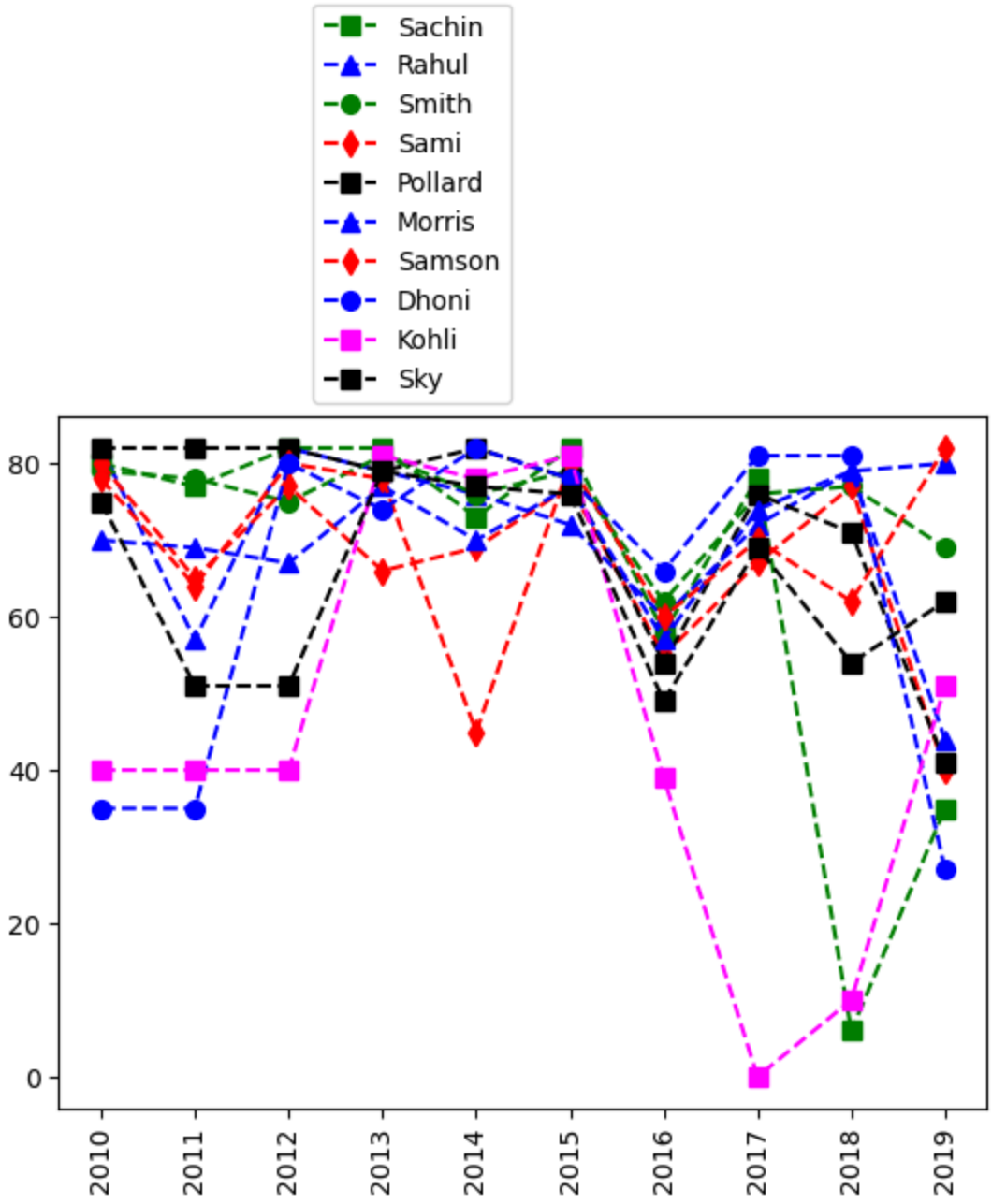
In [77]:
```
plt.plot(Salary[0], c='g', marker='s', ms=6, ls=':',label=Players[0])
plt.plot(Salary[1], c='b', marker='o', ms=8, ls=':',label=Players[1])
plt.plot(Salary[2], c='purple', marker='^', ms=8, ls='-',label=Players[2])
plt.plot(Salary[3], c='r', marker='d', ms=6, ls='-.',label=Players[3])
plt.legend(loc='upper left',bbox_to_anchor=(1,0))
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```



In [78]:
```
plt.plot(Salary[0], c='g', marker='s', ms=6, ls=':',label=Players[0])
plt.plot(Salary[1], c='b', marker='o', ms=8, ls=':',label=Players[1])
plt.plot(Salary[2], c='purple', marker='^', ms=8, ls='-',label=Players[2])
plt.plot(Salary[3], c='r', marker='d', ms=6, ls='-.',label=Players[3])
plt.legend(loc='lower right',bbox_to_anchor=(1,0))
plt.xticks(list(range(0,10)),Seasons)
plt.show()
```



In [86]:
```
plt.plot(Games[0], c='Green', marker='s', ms=7, ls='--',label=Players[0])
plt.plot(Games[1], c='Blue', marker='^', ms=7, ls='--',label=Players[1])
plt.plot(Games[2], c='Green', marker='o', ms=7, ls='--',label=Players[2])
plt.plot(Games[3], c='Red', marker='d', ms=7, ls='--',label=Players[3])
plt.plot(Games[4], c='Black', marker='s', ms=7, ls='--',label=Players[4])
plt.plot(Games[5], c='Blue', marker='^', ms=7, ls='--',label=Players[5])
plt.plot(Games[6], c='Red', marker='d', ms=7, ls='--',label=Players[6])
plt.plot(Games[7], c='Blue', marker='o', ms=7, ls='--',label=Players[7])
plt.plot(Games[8], c='Magenta', marker='s', ms=7, ls='--',label=Players[8])
plt.plot(Games[9], c='Black', marker='s', ms=7, ls='--',label=Players[9])
plt.legend(loc='lower right',bbox_to_anchor=(0.5,1))
plt.xticks(list(range(0,10)),Seasons,rotation='vertical')
plt.show()
```


Sachin
Rahul
Smith
Sami
Pollard
Morris
Samson
Dhoni
Kohli
Sky
80
60
40
20
0
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019

In [ ]: